

Vision- Link AI HUB



VISION-LINK
— AI HUB —

AI-POWERED HEALTHCARE FOR EVERY COMMUNITY

GenomePilot AI



Autonomous CRISPR
Genomic Intelligence
Platform



Powered by IBM Bob

Problem Statement

- Slow genomic analysis turnaround

- Dependence on bioinformatics specialists

- Complex CRISPR mutation interpretation

- Fragmented genomic software systems

- Limited workflow automation

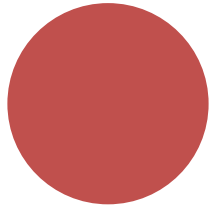
Our Solution

- GenomePilot AI is an autonomous AI-powered genomic intelligence platform that processes CRISPR genomic data end-to-end with AI agents and clinical reasoning.

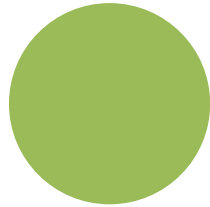
Why IBM Bob?

- • Repository understanding
- • API generation
- • Documentation automation
- • Test generation
- • Refactoring assistance
- • Workflow acceleration

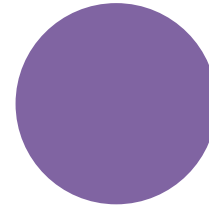
System Architecture



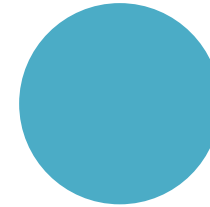
1. Data Ingestion
Layer



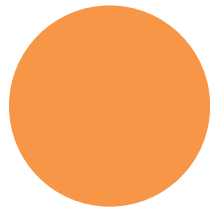
2. Genomic
Processing
Layer



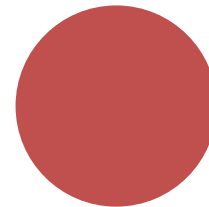
3. AI Agent
Orchestration



4. Knowledge
Retrieval

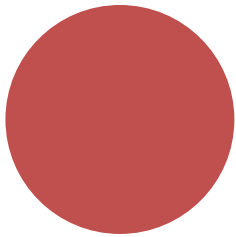


5. Clinical
Reasoning

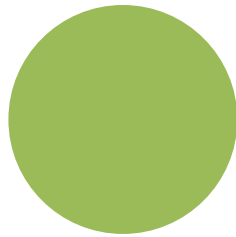


6. Backend
Infrastructure

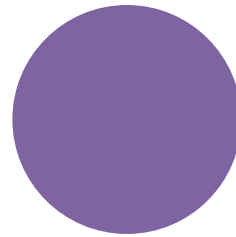
Data Ingestion Layer



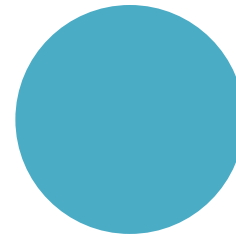
- FASTQ / BAM / VCF uploads



- Metadata validation



- Sequencing quality checks



- FastAPI + Cloud Storage

AI Agent Orchestration

- Task planning

- Workflow execution

- Error handling

- Multi-step reasoning

- LangGraph + CrewAI

Genomic Intelligence Engine

- Mutation detection
- Variant classification
- Disease risk prediction
- CRISPR off-target analysis

Knowledge Retrieval

- ClinVar

- NCBI

- PubMed

- Mutation
repositorie
s

- RAG-
based
intelligenc
e

Backend Infrastructure

- FastAPI
- PostgreSQL
- Celery + Redis
- Docker
- Kubernetes
- Prometheus + Grafana

Security & Compliance

- HIPAA compliance
- End-to-end encryption
- Role-based access control
- Secure audit logging

Autonomous Logic Flow

- Upload → Validation → Preprocessing → AI Analysis → Knowledge Retrieval → Clinical Reasoning → Report Generation

IBM Bob in Action

- Examples:
 - • Generate FastAPI endpoints
 - • Explain workflows
 - • Create tests
 - • Refactor services

Demo Flow

- 1. Upload genomic file
- 2. AI pipeline activates
- 3. Mutation analysis runs
- 4. Clinical report generated
- 5. IBM Bob assists development

Enterprise Benefits

- • Faster genomic analysis
- • Reduced operational costs
- • Better treatment planning
- • Enterprise scalability

Future Scope

- • Real-time genomic monitoring
- • AI-assisted drug discovery
- • Predictive oncology intelligence

Conclusion

- GenomePilot AI combines AI agents, genomic intelligence, RAG systems, and IBM Bob to accelerate precision healthcare.



Thank You

- GenomePilot AI
- Powered by IBM Bob
- Questions & Discussion